

Yuli Multi-Enhancing Artificial Vision by Integrating Machine Learning in Retinal Prosthetics

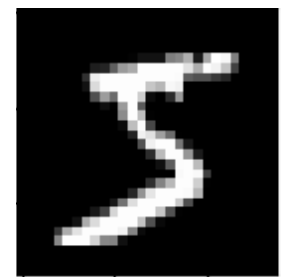
RWTHAACHEN
UNIVERSITY

1

6

End-to-End Encoding

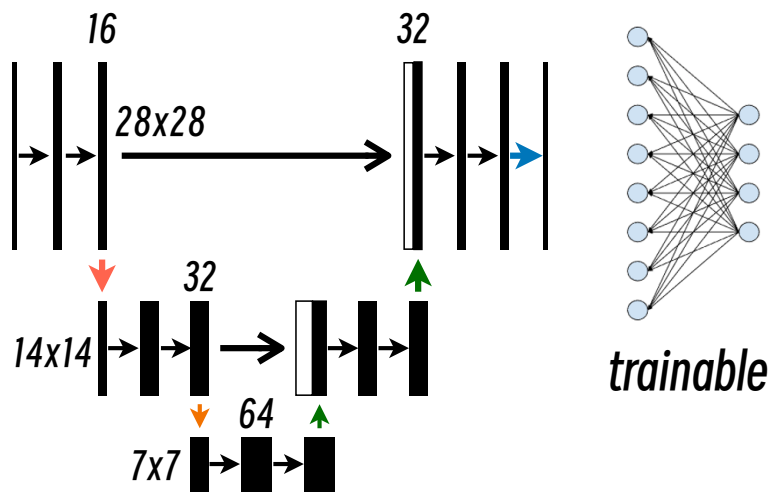
Input Image



28x28

{ '5' }

Downsampling



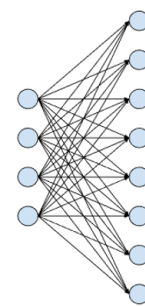
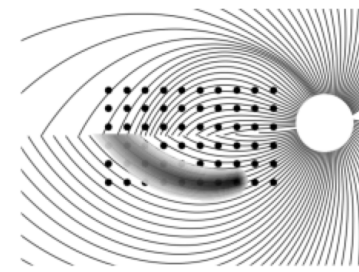
CNN Encoder

Stimulation Pattern



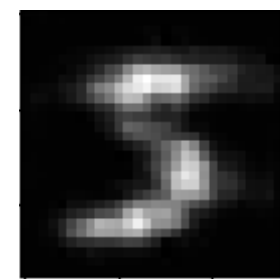
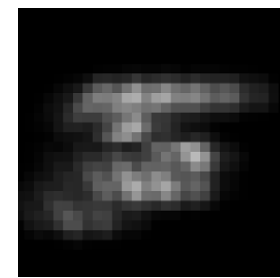
6x10
(or 28x28)

pulse2percept



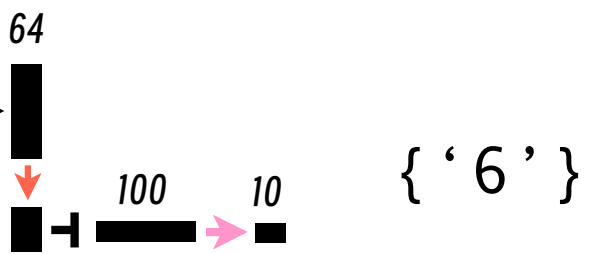
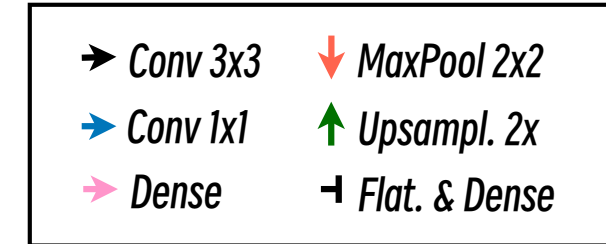
CNN Implant Model

Predicted Percept



28x28

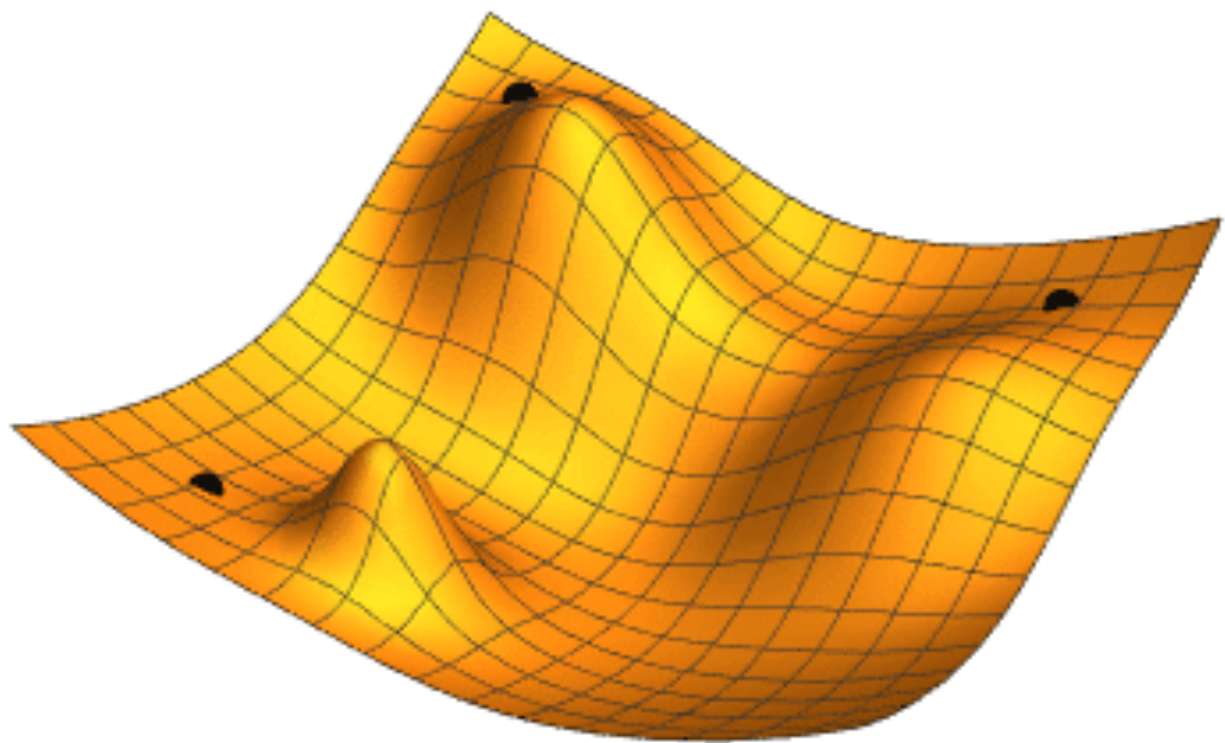
32



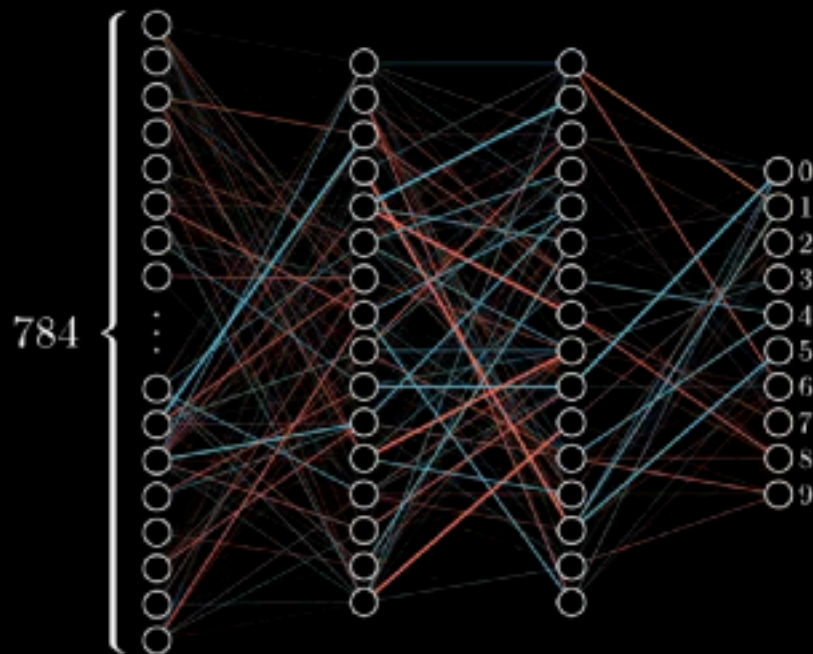
VGG-5
Classifier

{ '6' }

{ '5' }



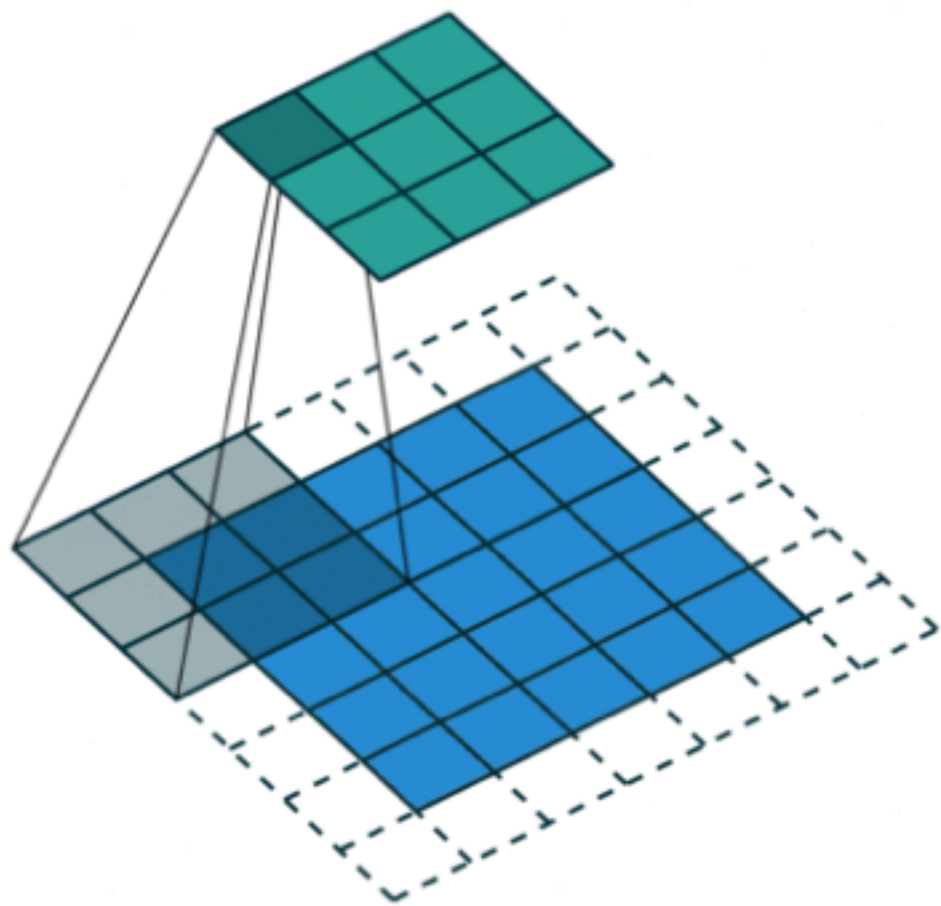
Training in
progress...





© 3Bue1Brown

Abiotic Donnan



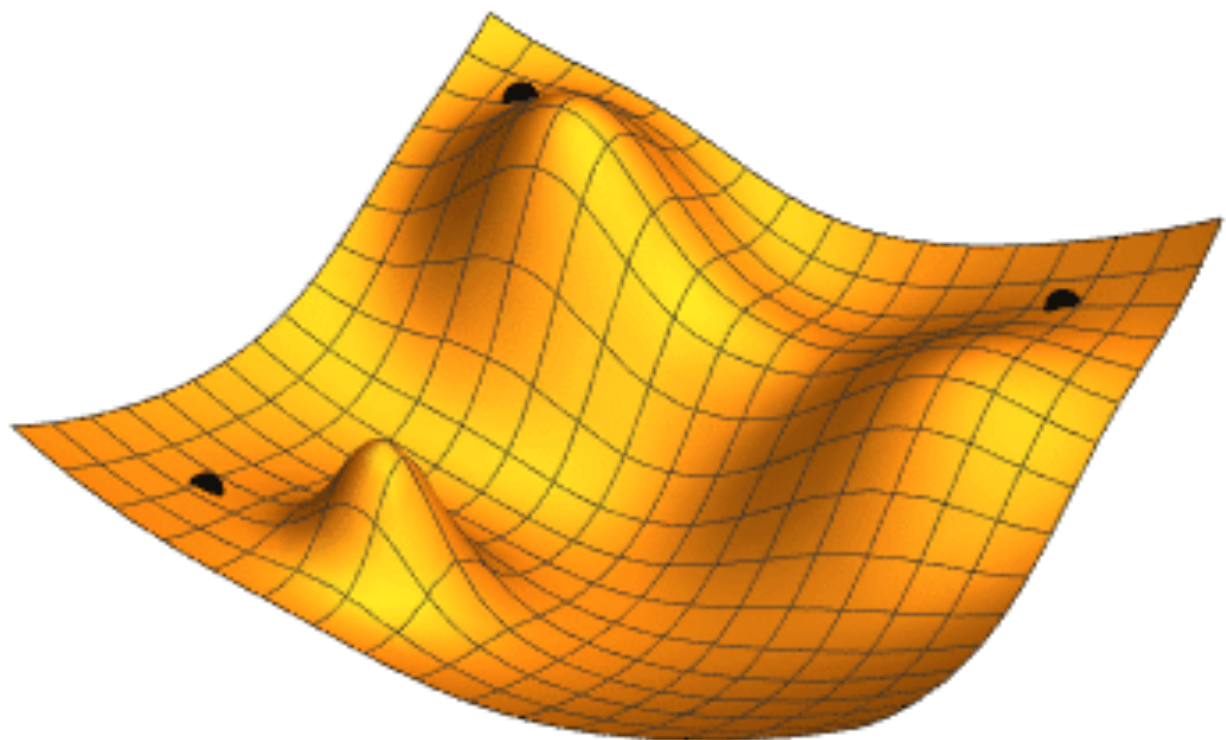
[Wuetal., 2023]

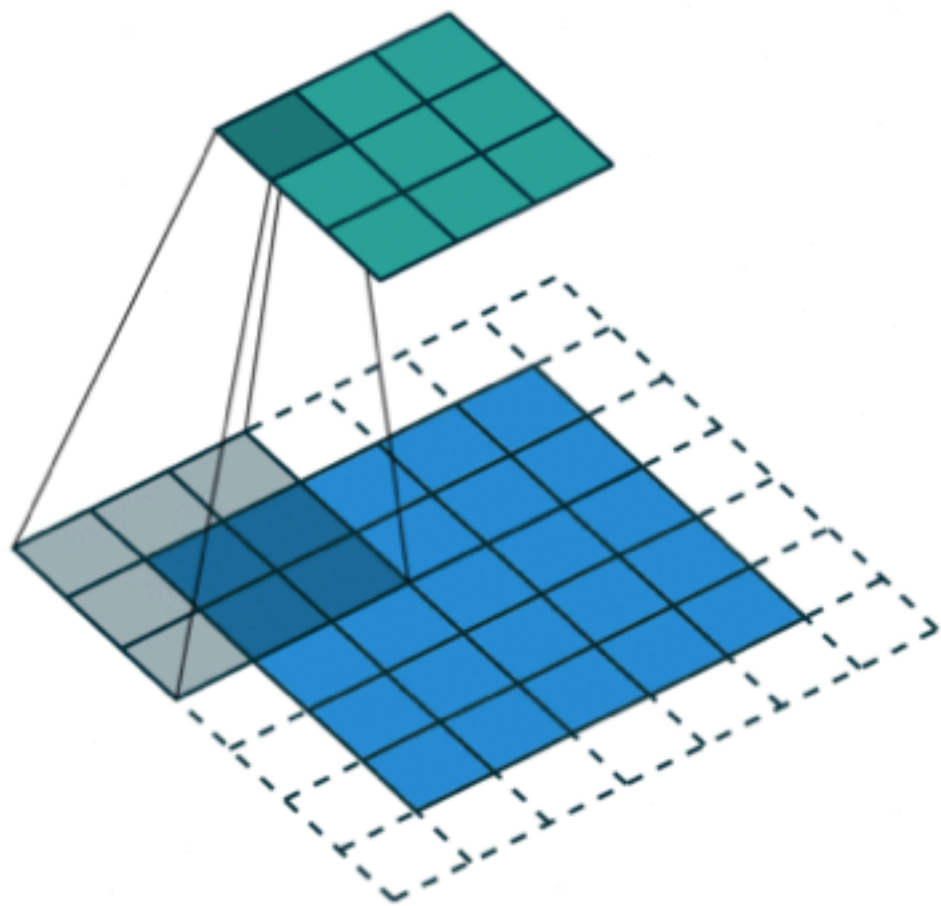
[Dumoulin & Visin, 2018]

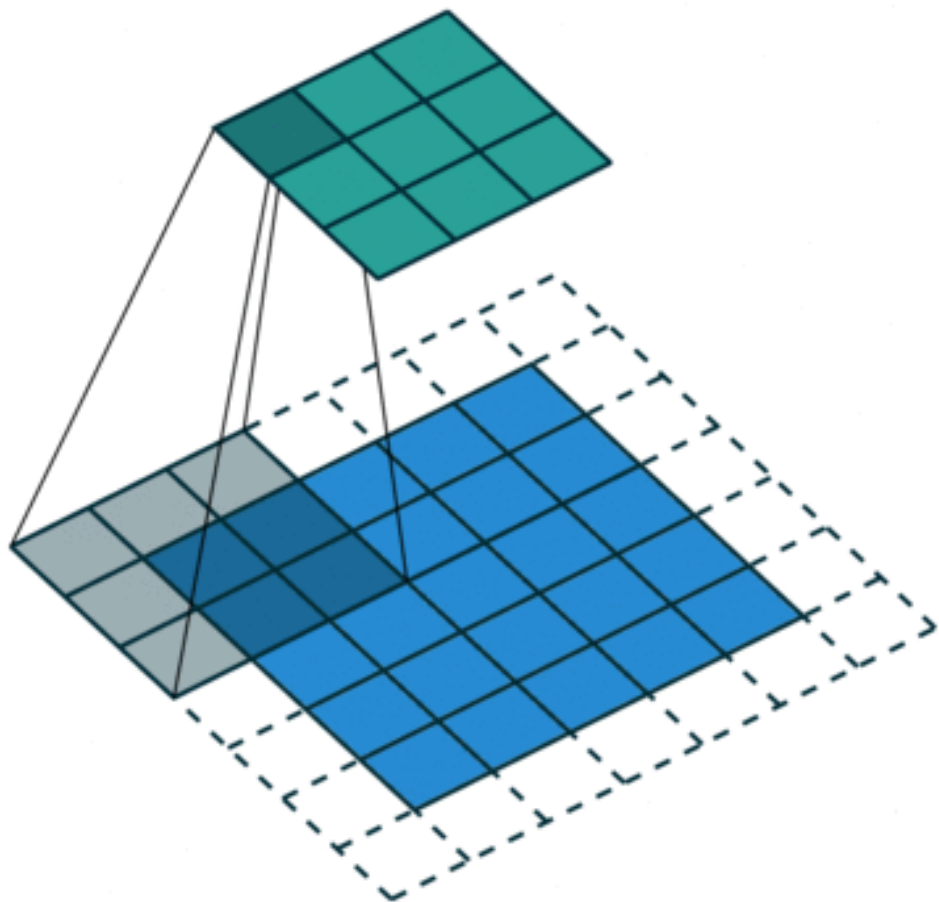
Convolutional Layer

Modeling

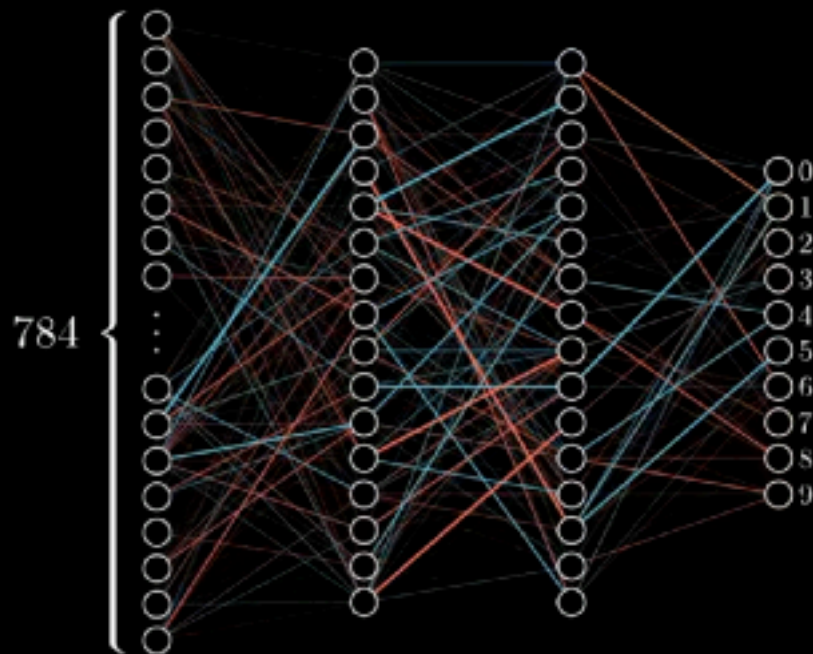
Grandient Descent



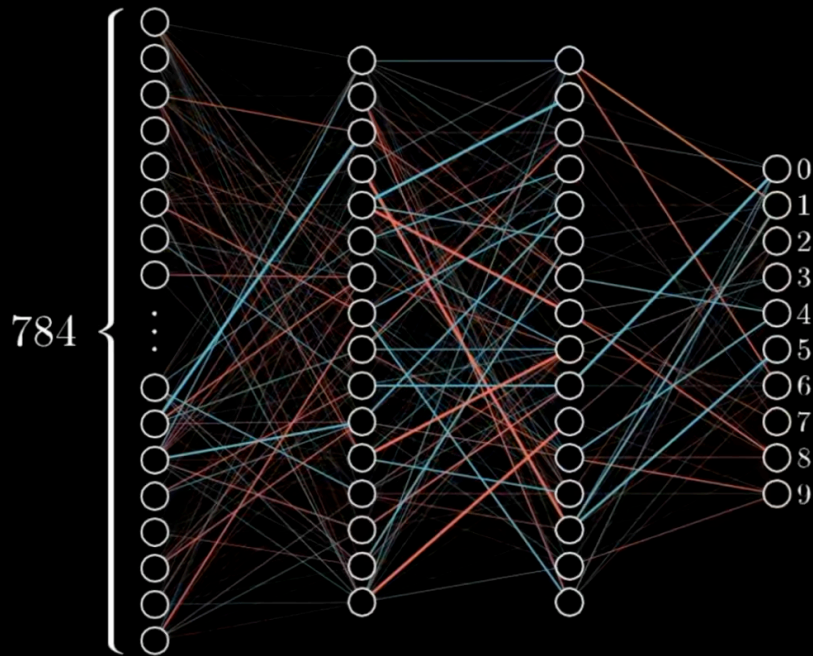




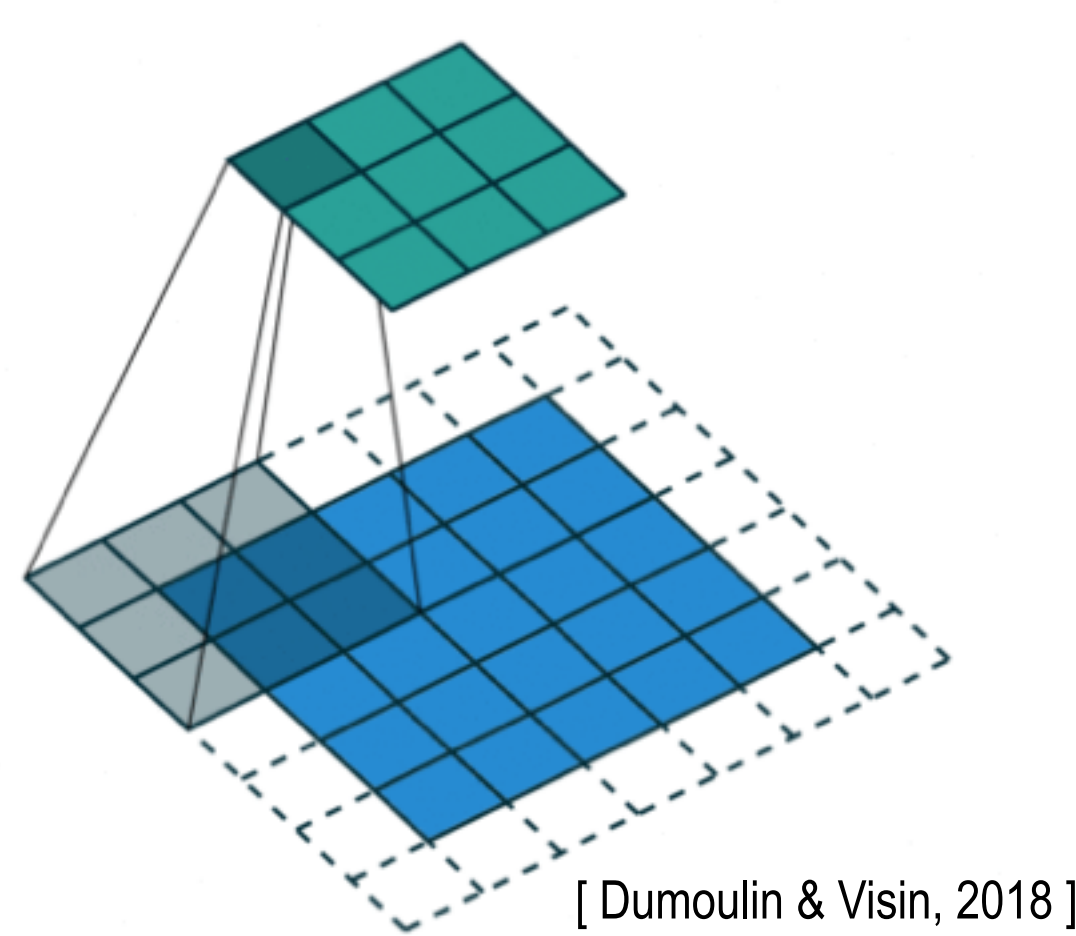
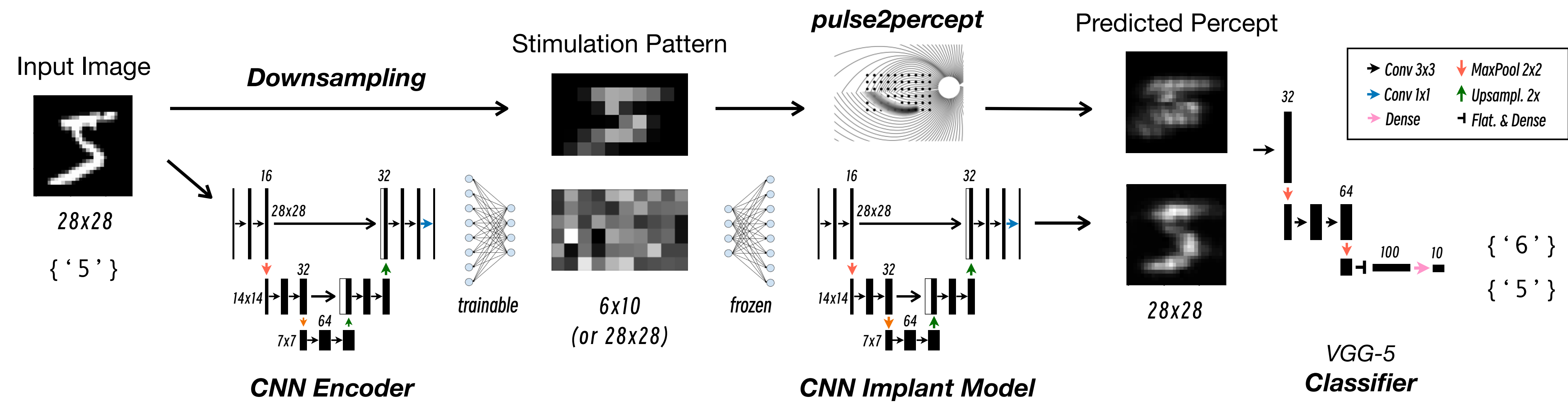
Training in
progress...



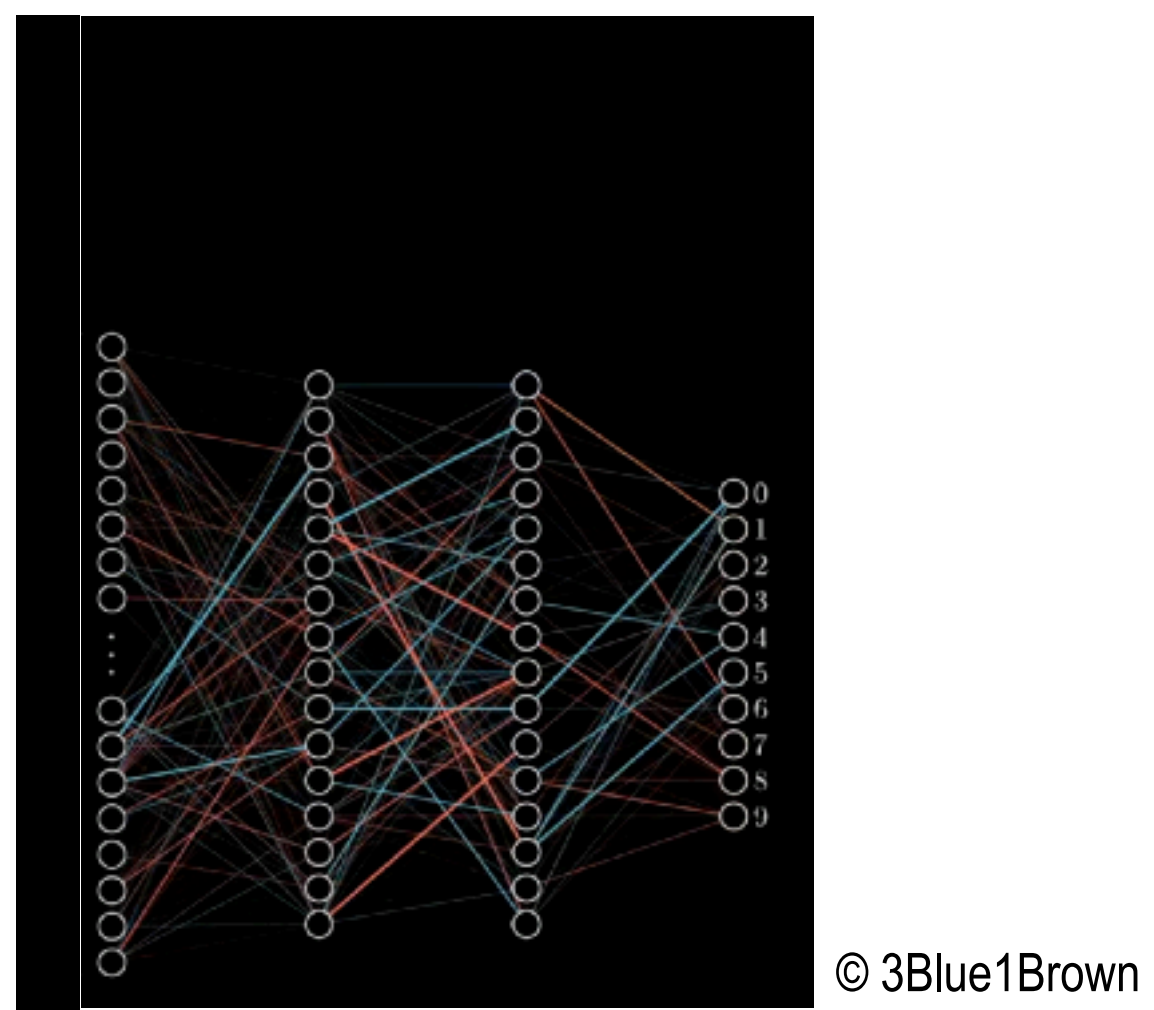
Training in
progress...



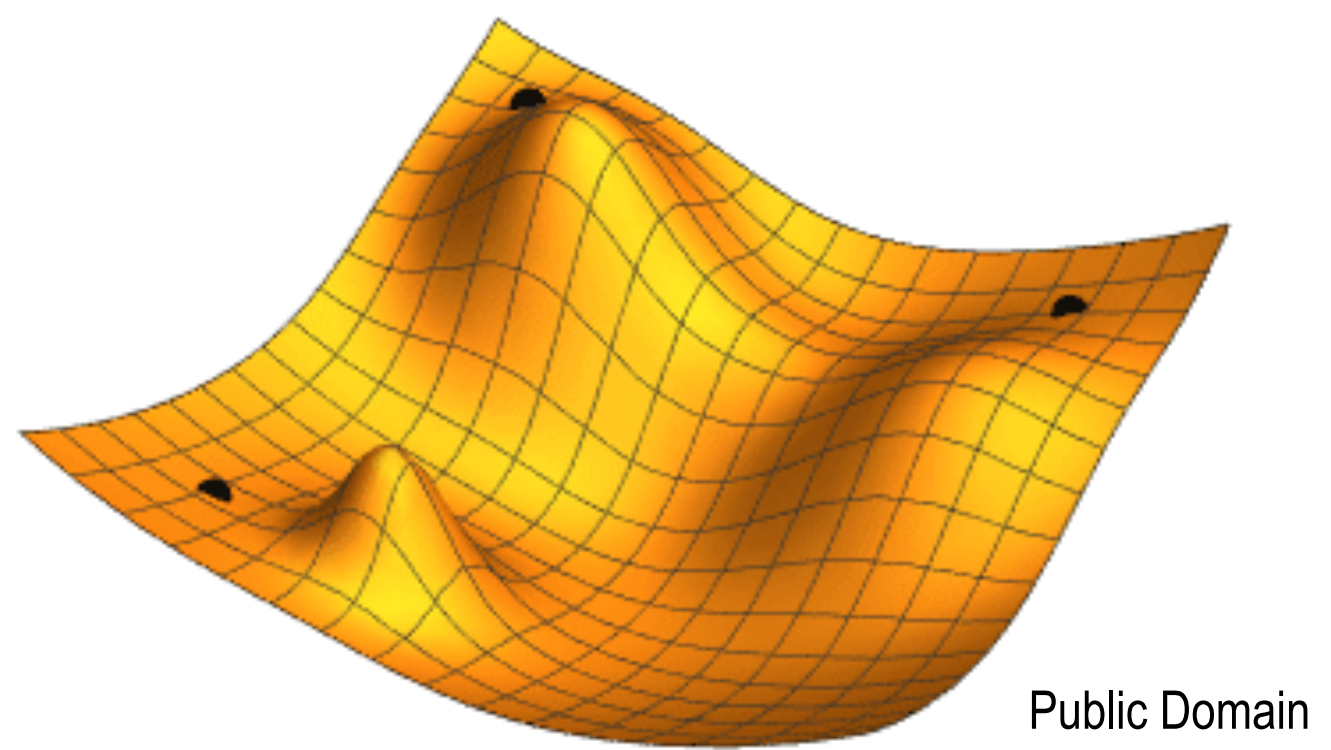
End-to-End Encoding



[Wu et al., 2023b] Convolutional Layer



Model Updating

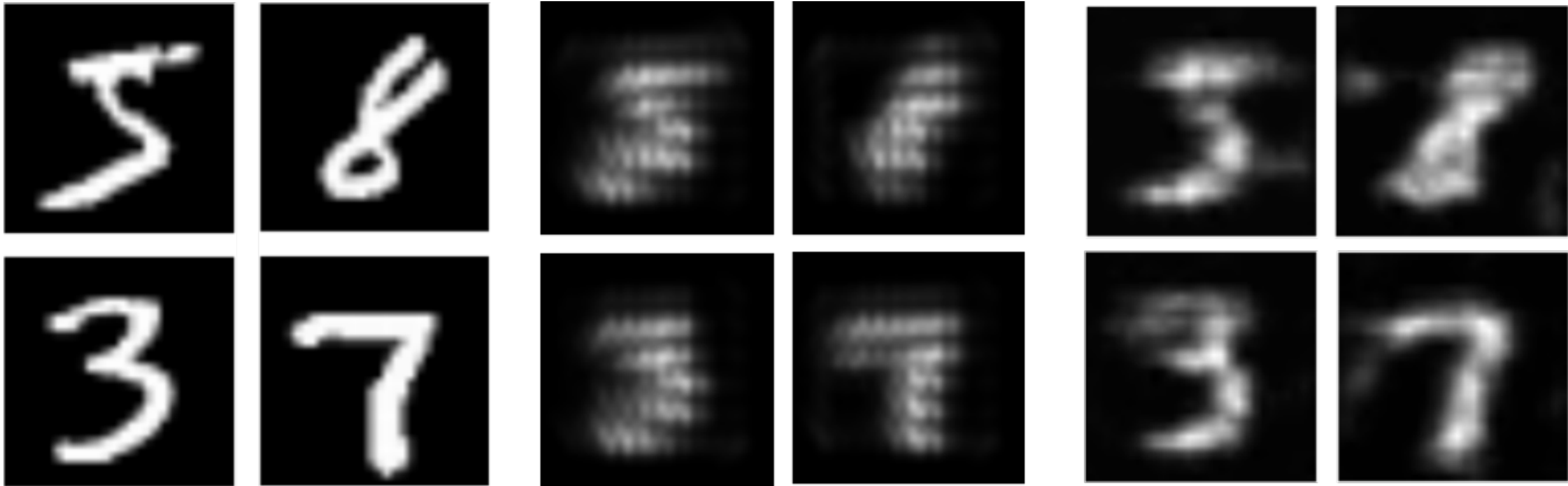


Gradient Descent

End-to-End Encoding

Stim. Res.	Encoder	Loss	Micro F1	Weighted F1
28 × 28	None	-	77.28%	76.63%
28 × 28	U-Net	MSE	98.08%	98.07%
28 × 28	U-Net	CE	98.81%	98.81%
6 × 10	Downsampled	-	61.87%	60.68%
6 × 10	U-Net	MSE	86.37%	85.78%
6 × 10	U-Net	CE	96.84%	96.85%

+36%

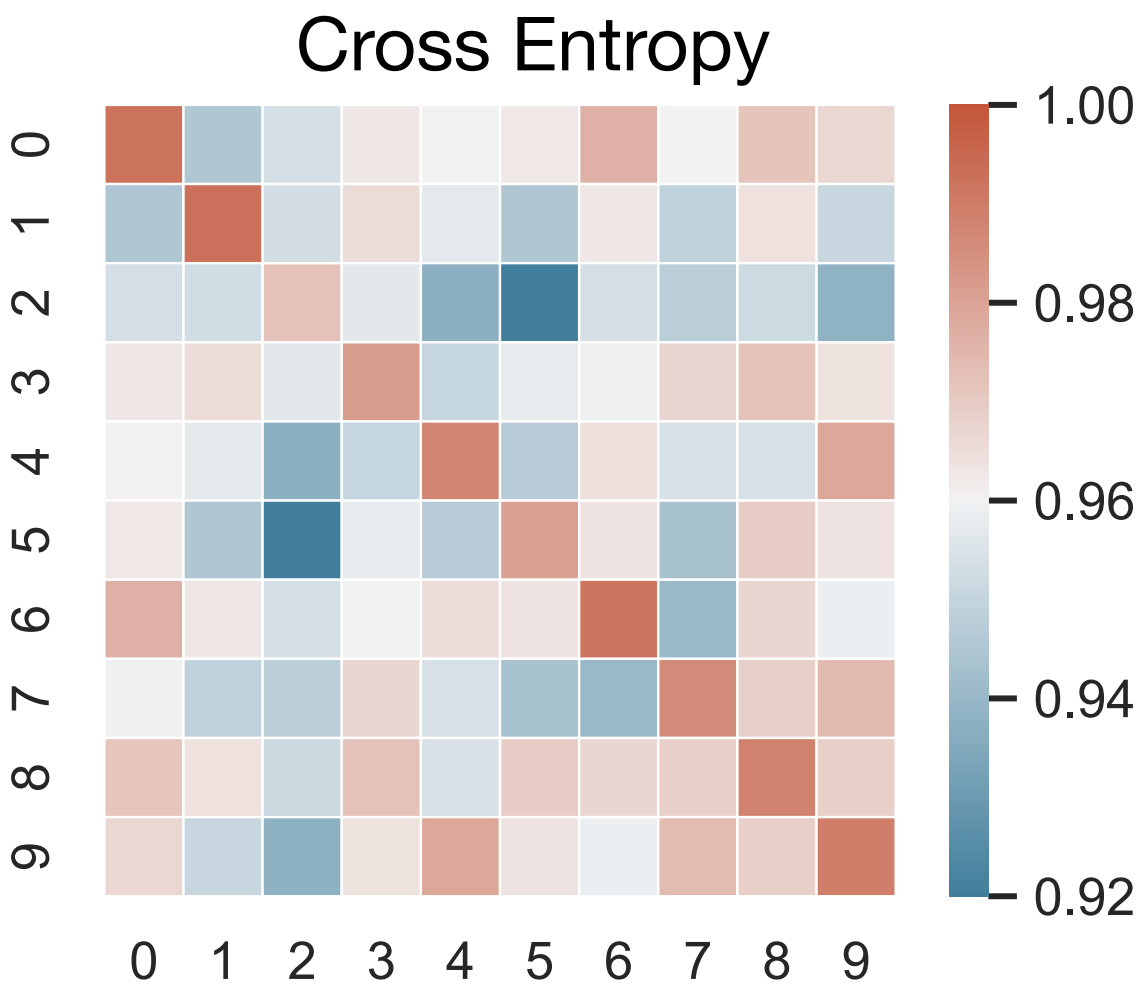
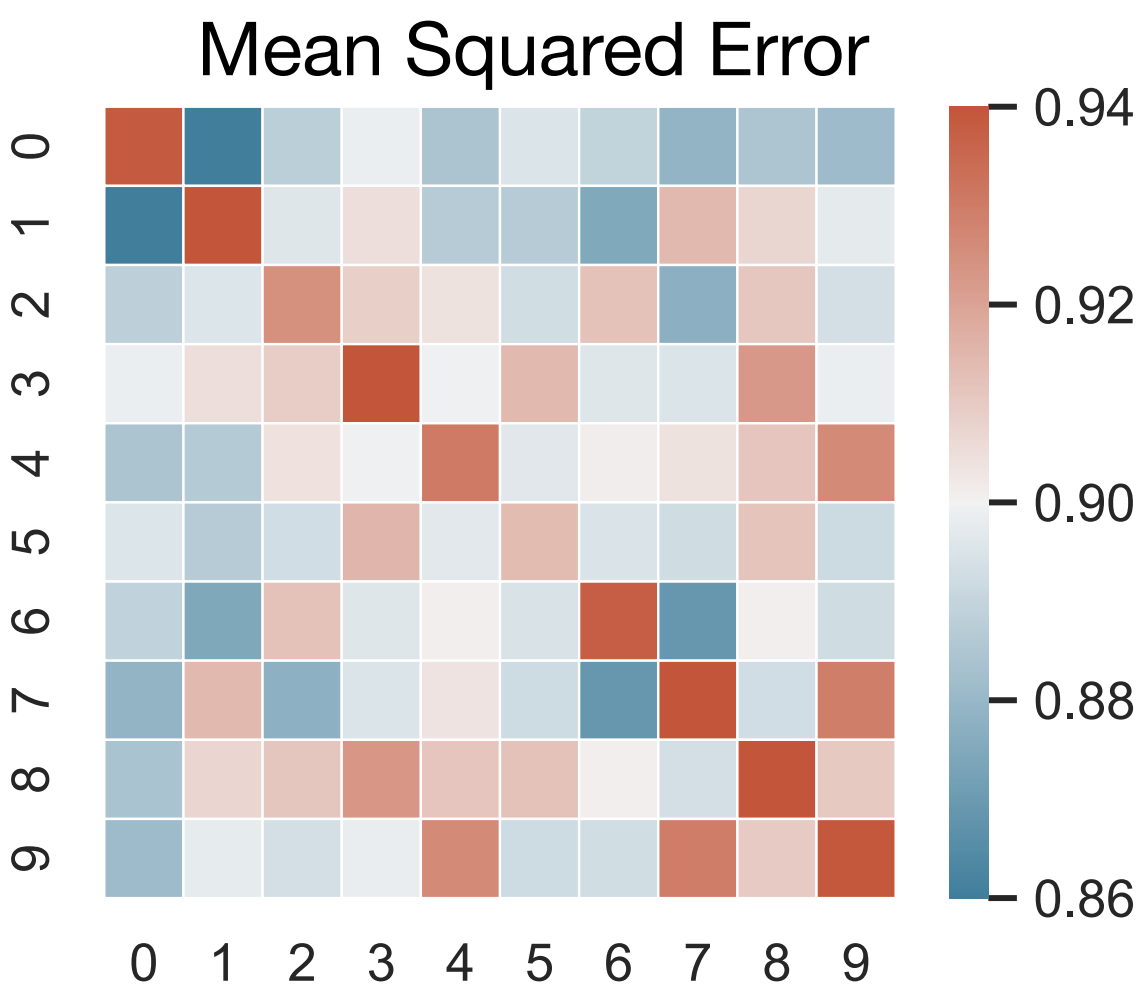


Original Images

Percepts with Downsampling

Percepts with End-to-end Encoding

[Wu et al., 2023b]



Cosine Similarity between Stimuli